Microsoft

WorkshopPLUS: Azure AI Platform and Services

Azure AI Vision

Conditions and terms of use

Learning Units covered in this Module

- What is computer vision?
- Optical character recognition
- Azure AI Image analysis
- Azure AI custom vision
- Azure AI face service
- Azure AI video indexer

Objectives

After completing this Learning, you will be able to:

1

Understand the Fundamentals of Computer Vision
- ✓ Learn what computer vision is and explore its core concepts and applications.

2

Understand and Apply Optical Character Recognition (OCR)
- ✓ Understand how OCR works, including printed and handwritten text extraction, and its deployment options (cloud and on-premises).

3

Utilize Azure AI Image Analysis
- ✓ Extract visual features from images, such as content tags, captions, object detection, and smart cropping.
- ✓ Recognize products, people, and common objects in images.

4

Leverage Azure AI Custom Vision
- ✓ Build, deploy, and improve custom image classification and object detection models tailored to specific needs.
- ✓ Understand the different domains available for classification and detection.

5

Explore Azure AI Face Service
- ✓ Detect and analyze facial features, head pose, and other attributes.
- ✓ Understand responsible AI principles and limited access policies for facial recognition.

6

Explore Azure AI Face Service
- ✓ Extract insights from video content, including face detection, OCR, content moderation, and scene segmentation.
- ✓ Apply video and audio models for tasks like transcription, speaker identification, and sentiment analysis.

What is Computer Vision?

A Glimpse of Diverse AI Vision Tasks

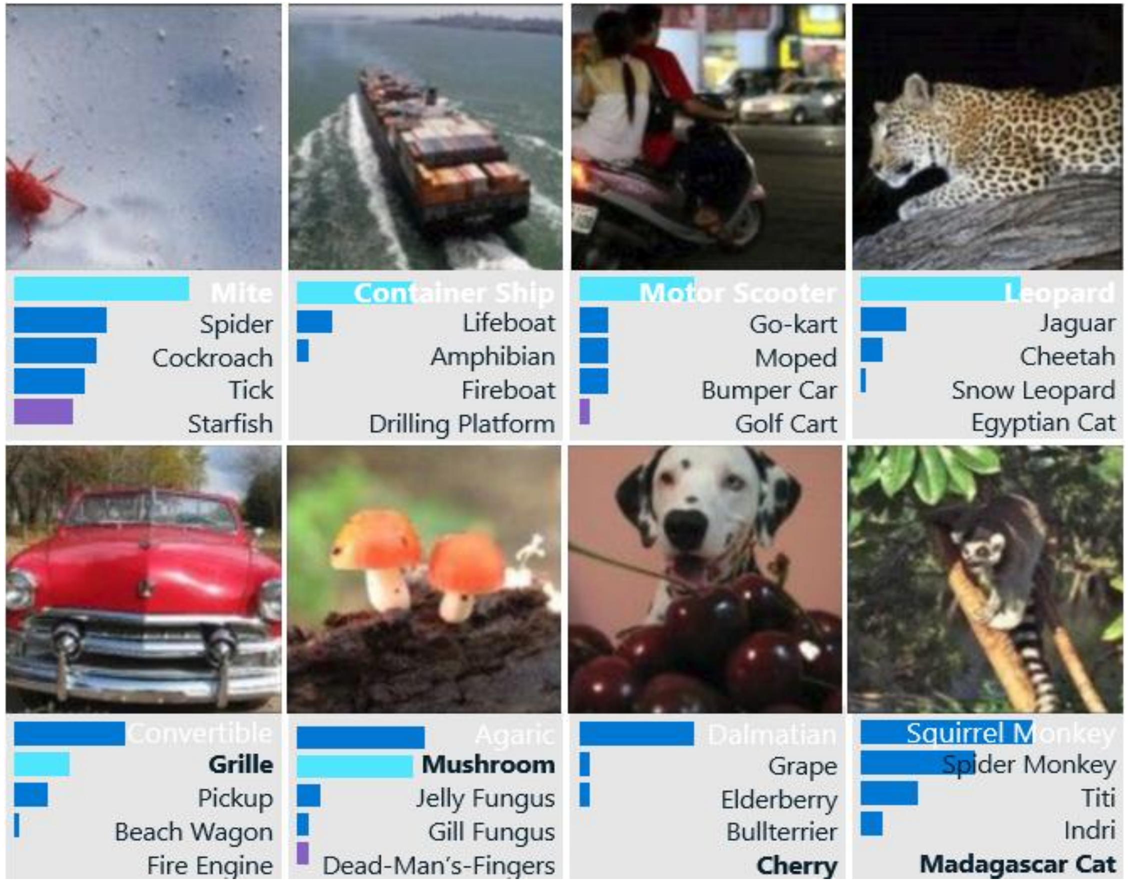


Image classification

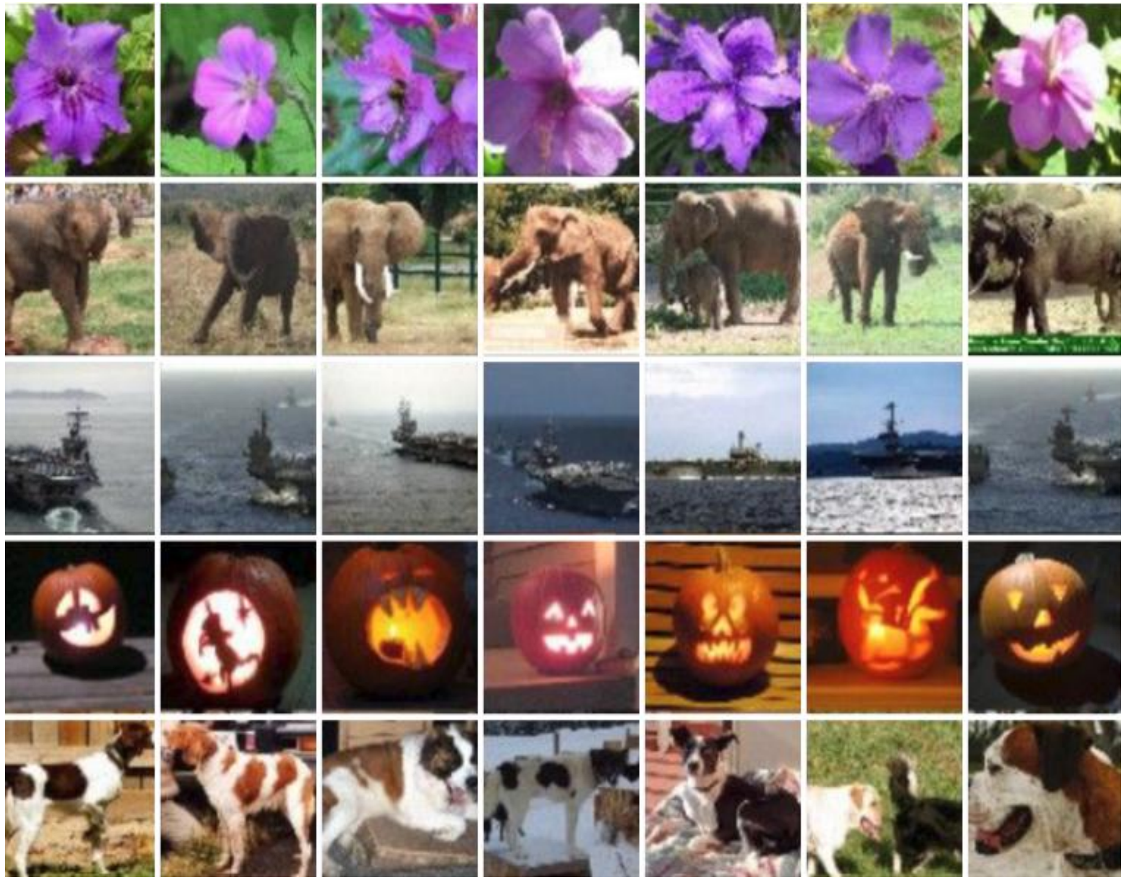
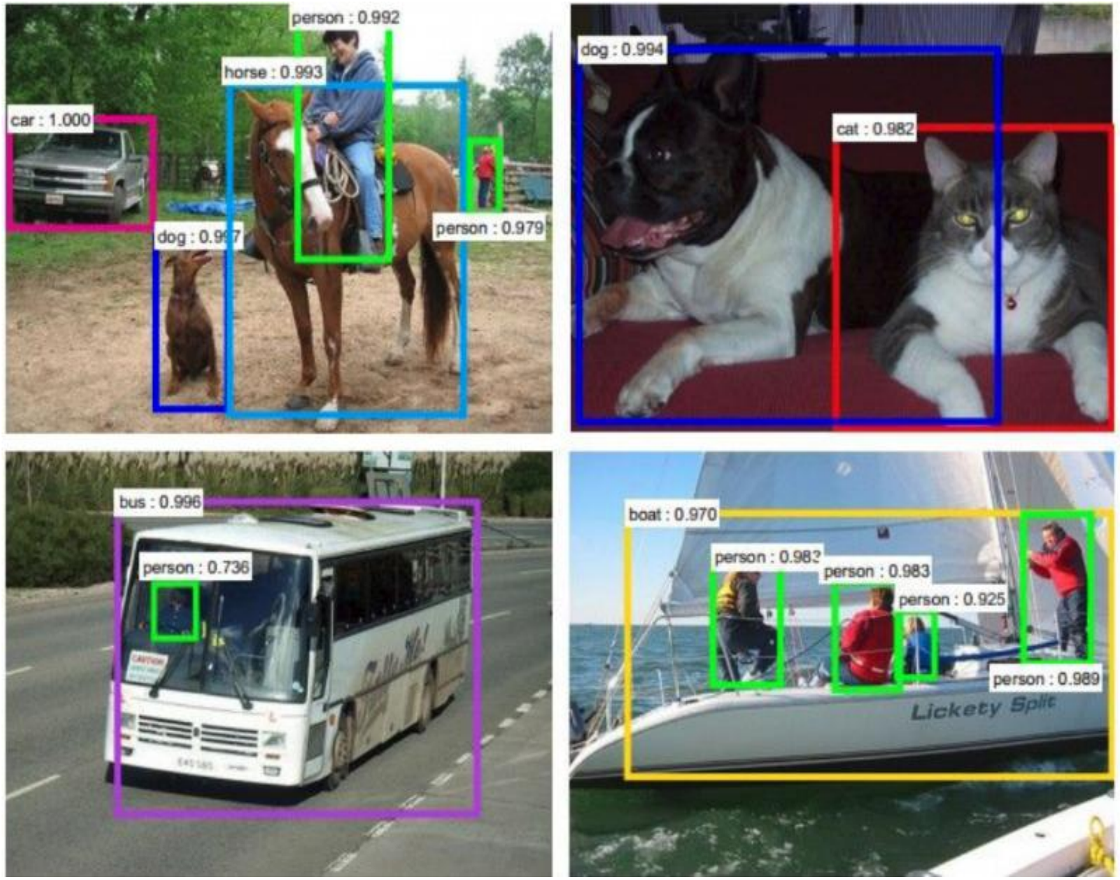


Image retrieval

Krizhevsky, Sutskever, and Hinton, 2012

A Glimpse of Diverse AI Vision Tasks



Object detection
Ren, He, Girshick, and Sun 2015

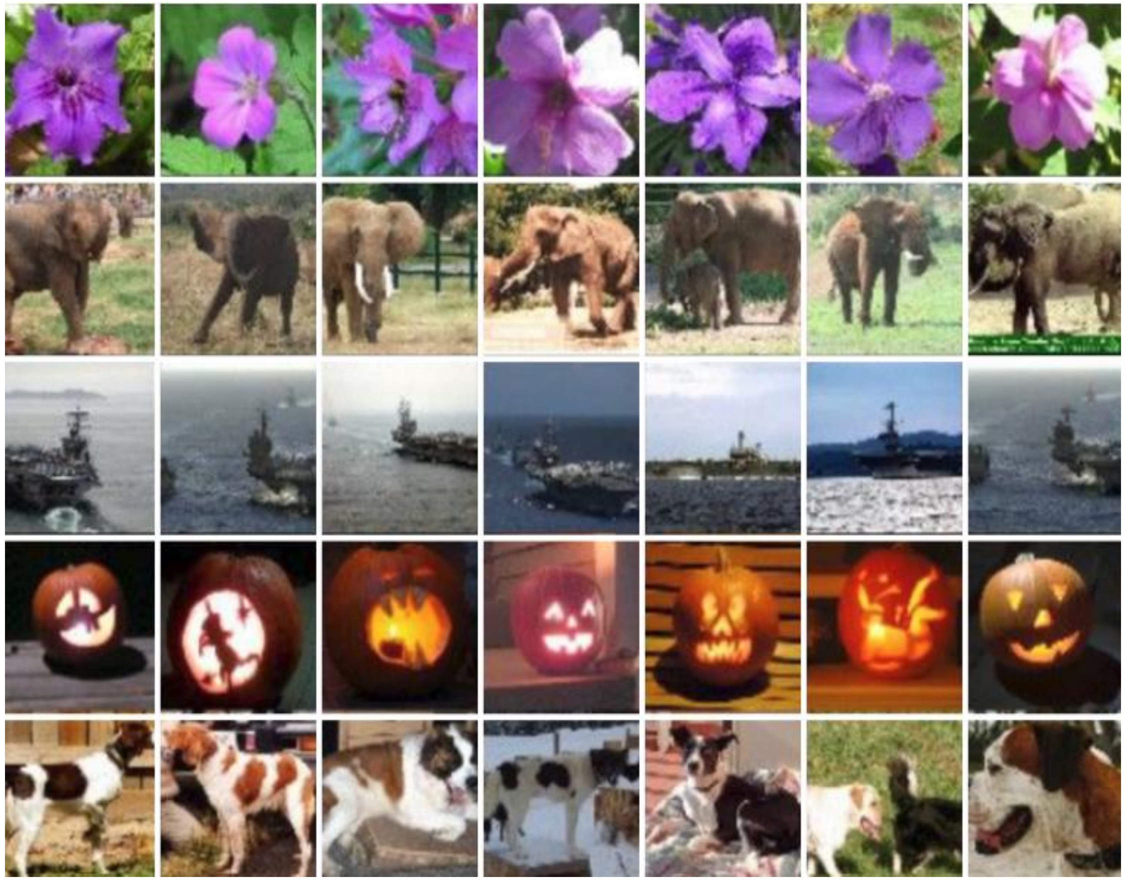
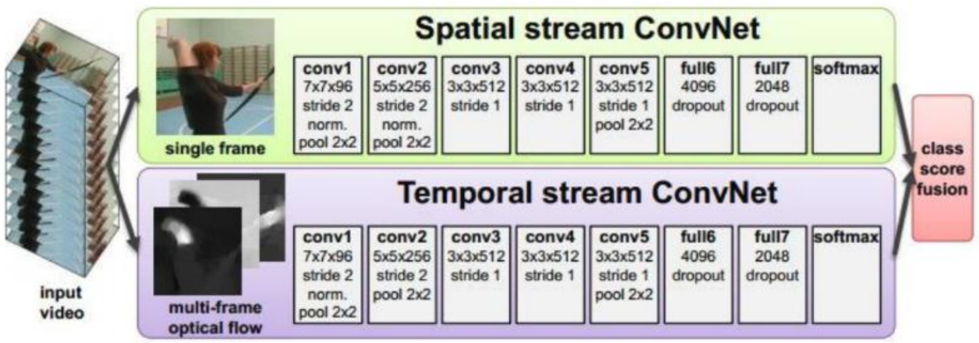


Image segmentation
Fabaret et al, 2012

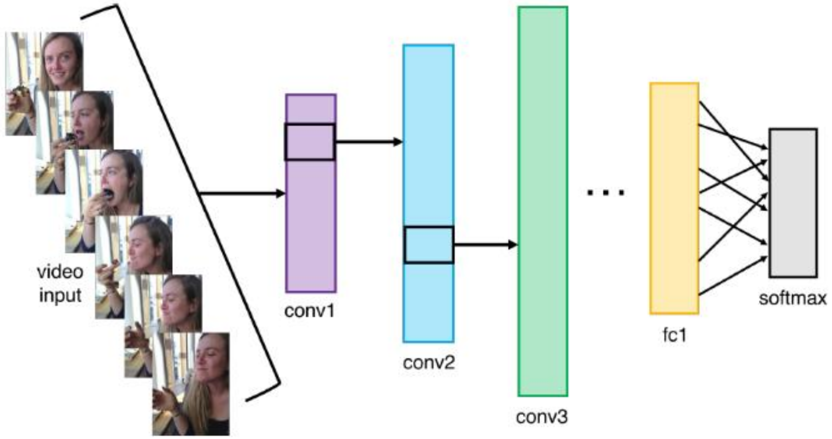
A Glimpse of Diverse AI Vision Tasks

Video Classification



Simonyan et al, 2014

Activity Recognition



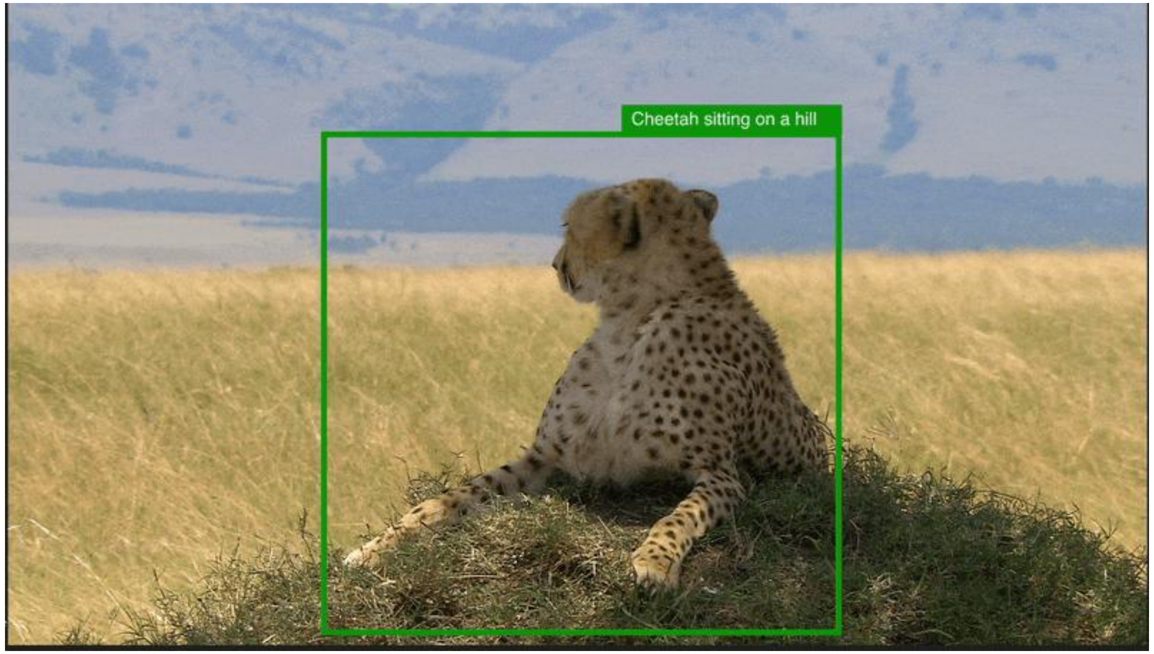
Pose Recognition (Toshev and Szegedy, 2014)

Azure AI Vision

Extract robust insights from images and video content across multiple industry domains

With Azure AI Vision, developers can create cutting-edge, market-ready, responsible computer vision applications that seamlessly digitize, analyze, and connect data to natural language interactions. And they can customize the model with few-shot learning.

- Boost content discoverability with natural language
- Support accessibility
- Drive acquisition through SEO
- Enable real-time alerts
- Enhance security and safety
- Improve incident response times
- Enable superior user experience through content-based image retrieval
- Fine-tune with few-shot learning



Features of Azure AI Vision

FEATURES

Analyze visual content in different ways with Azure AI Vision

Image analysis

Image analysis that pulls from more than 10,000 concepts and objects to detect, classify, caption, and generate insights.

Spatial analysis

Spatial analysis to understand people's presence and movements within physical areas in real time.

Optical character recognition (OCR)

Optical character recognition (OCR) to extract printed and handwritten text from images with varied languages and writing styles.

Facial recognition

Facial recognition to create intelligent applications that recognize and verify human identity.

Optical Character Recognition (OCR)

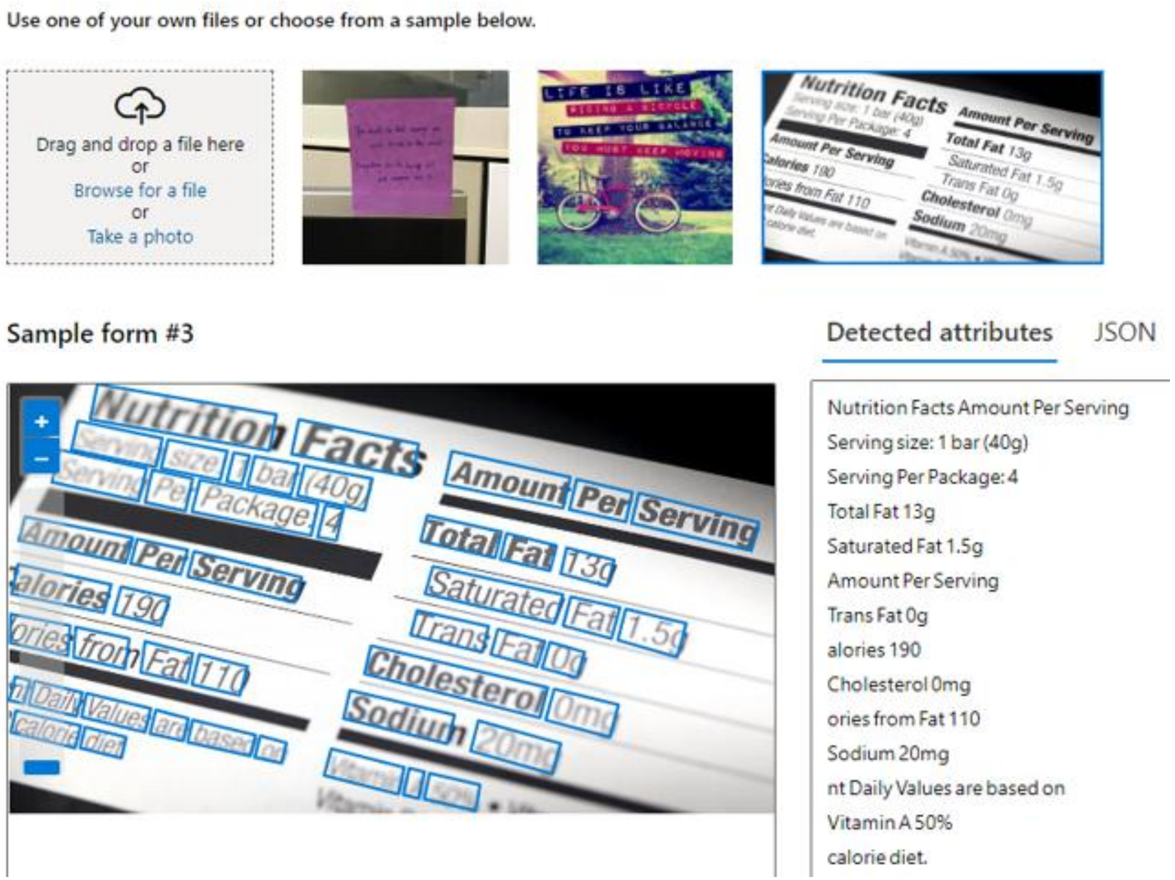
OCR – Optical Character Recognition

Microsoft's **Read** OCR engine is composed of multiple advanced machine-learning based models supporting global languages.

It can extract printed and handwritten text including mixed languages and writing styles.

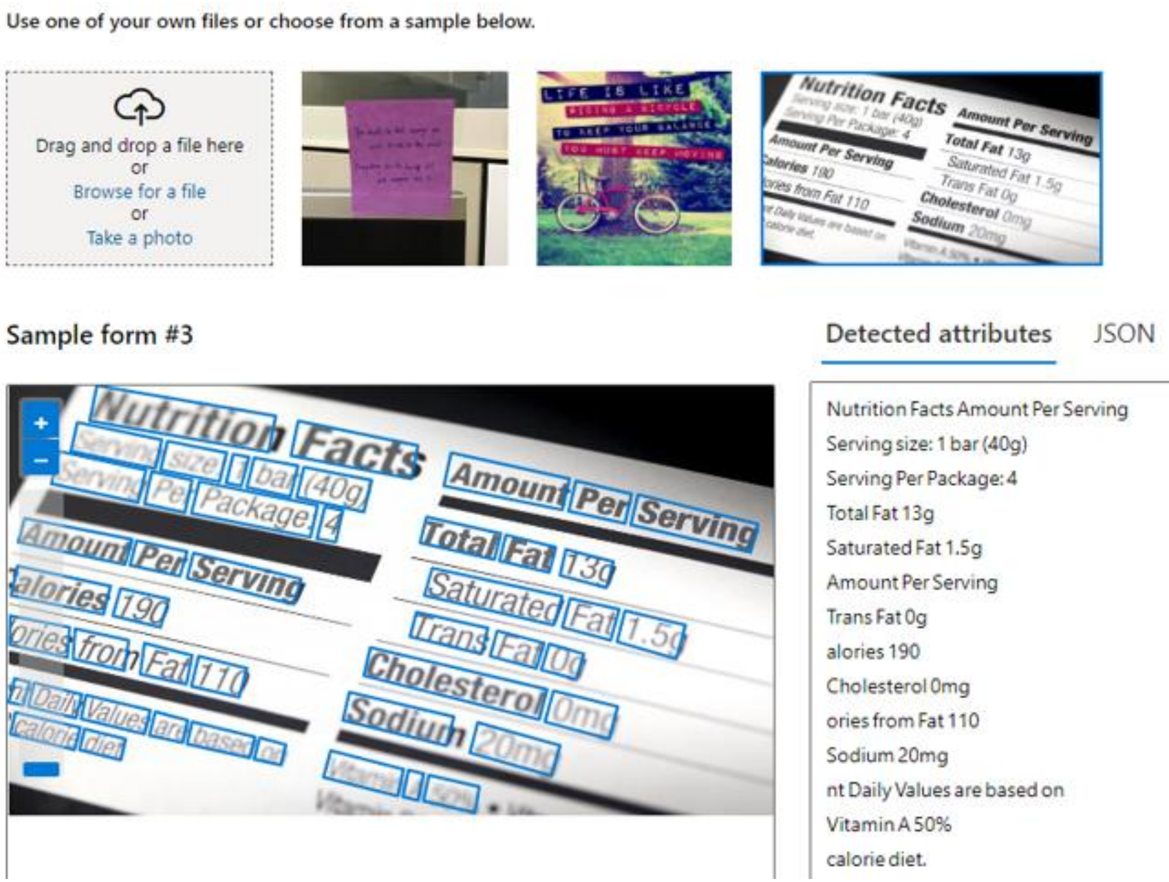
Read is available as cloud service and on-premises container for deployment flexibility.

It's also available as a synchronous API for single, non-document, image-only scenarios.



Common Features of Read OCR

- Printed and handwritten text extraction in supported languages
- Pages, text lines and words with location and confidence scores
- Support for mixed languages, mixed mode (print and handwritten)
- Available as Distroless Docker container for on-premises deployment

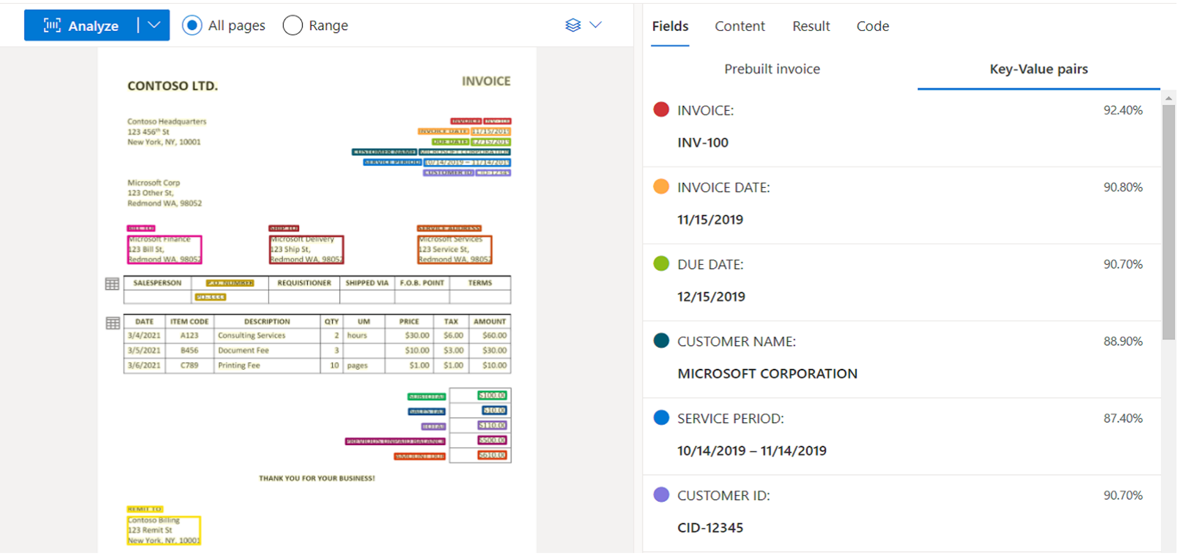


IDP – Intelligent Document Processing

- Intelligent Document Processing (IDP) uses OCR as its foundational technology to additionally extract structure, relationships, key-values, entities, and other document-centric insights

- Azure AI Document Intelligence includes a document-optimized version of **Read** as its OCR engine.

- Ideal for scenarios of extracting text from scanned and digital documents



Using OCR On-premises

Containers let you run Azure AI Vision APIs in your own environment.

Help the customers to meet specific security and data governance requirements.

The host computer can be:

- X64-based computer on premises or
- Docker hosting service in Azure (Azure Kubernetes Service, Azure Container Instance, Kubernetes cluster deployed to Azure stack)

Reference: Azure AI Vision 3.2 GA Read OCR container - Azure AI services | Microsoft Learn

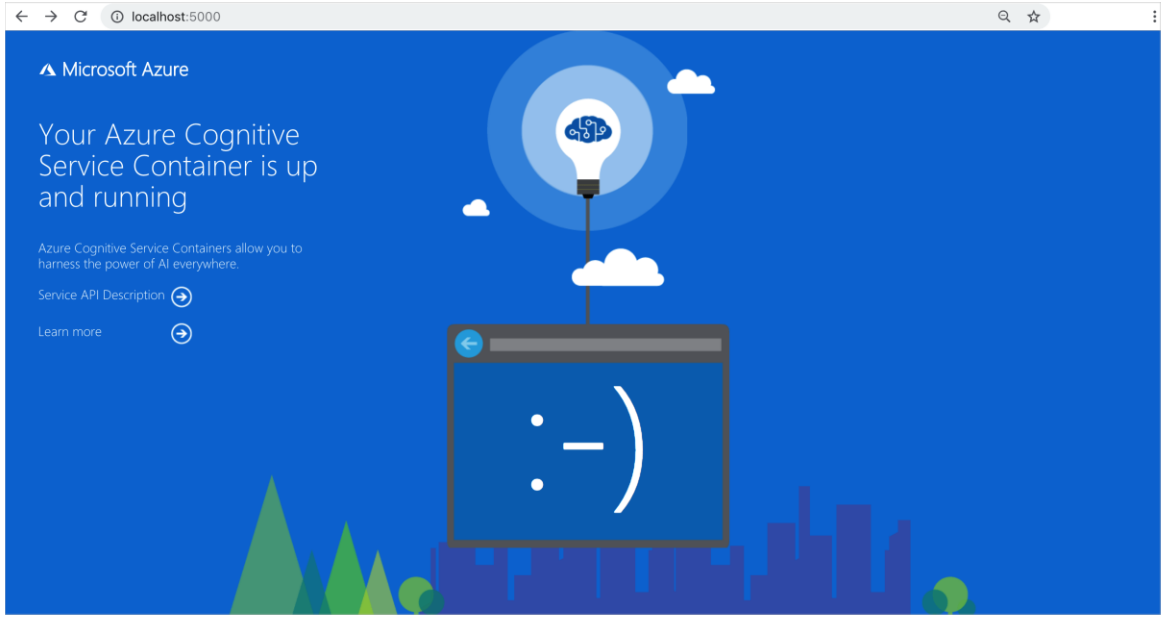
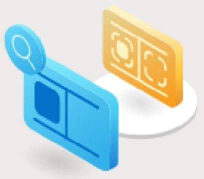


Image Analysis

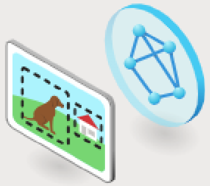
Image Analysis Capabilites



Recognize products on shelves
Preview

Identify products on shelves, gaps in product availability, and compliance for planograms.

Try it out



Customize models with images
Preview

Create custom image classification and object detection models with images using Vision Studio and Azure ML.

Start a project

Search photos with image retrieval

Retrieve specific moments within your photo album. For example, you can search for: a wedding you attended last summer, your pet, or your favorite city.

Try it out

Add dense captions to images

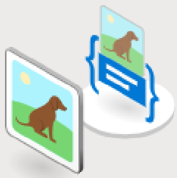
Generate human-readable captions for all important objects detected in your image.

Try it out

Remove backgrounds from images
Preview

Easily remove the background and preserve foreground elements in your image.

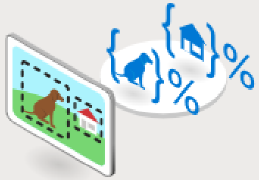
Try it out



Add captions to images

Generate a human-readable sentence that describes the content of an image.

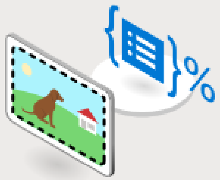
Try it out



Detect common objects in images

Recognize the location of objects of interest in an image and assign them a label.

Try it out



Extract common tags from images

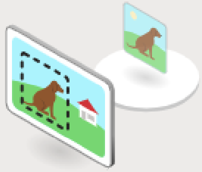
Use an AI model to automatically assign one or more labels to an image.

Try it out

Detect sensitive content in images

Detect sensitive content in images so you can moderate their usage in your applications.

Try it out



Create smart-cropped images

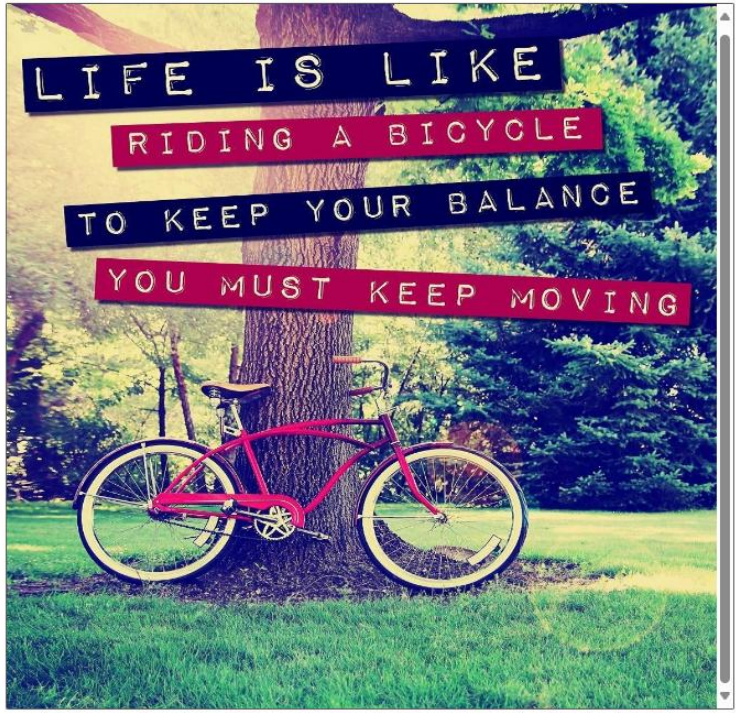
Create cropped image thumbnails based on the key areas of a larger image.

Try it out

Extract Common Tags from Images

Image Analysis can return content tags for thousands of recognizable objects, living beings, scenery, and actions that appear in images.

Sample image 4

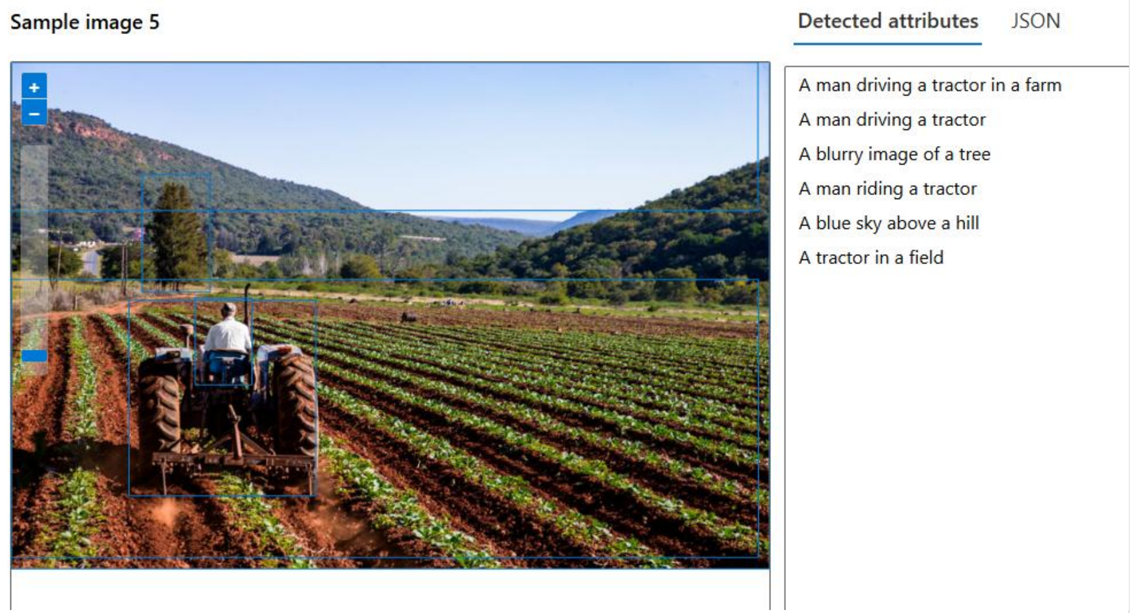


Detected attributes JSON

text (99.99%)
wheel (98.89%)
outdoor (98.69%)
bicycle wheel (98.58%)
bike (97.93%)
land vehicle (97.03%)
grass (95.55%)
sign (94.98%)
vehicle (94.69%)
bicycle (94.68%)
bicycle frame (89.65%)
bicycle tire (89.65%)
tire (89.55%)
transport (89.16%)
cycling (87.18%)
bicycles--equipment and supplies (85.49%)

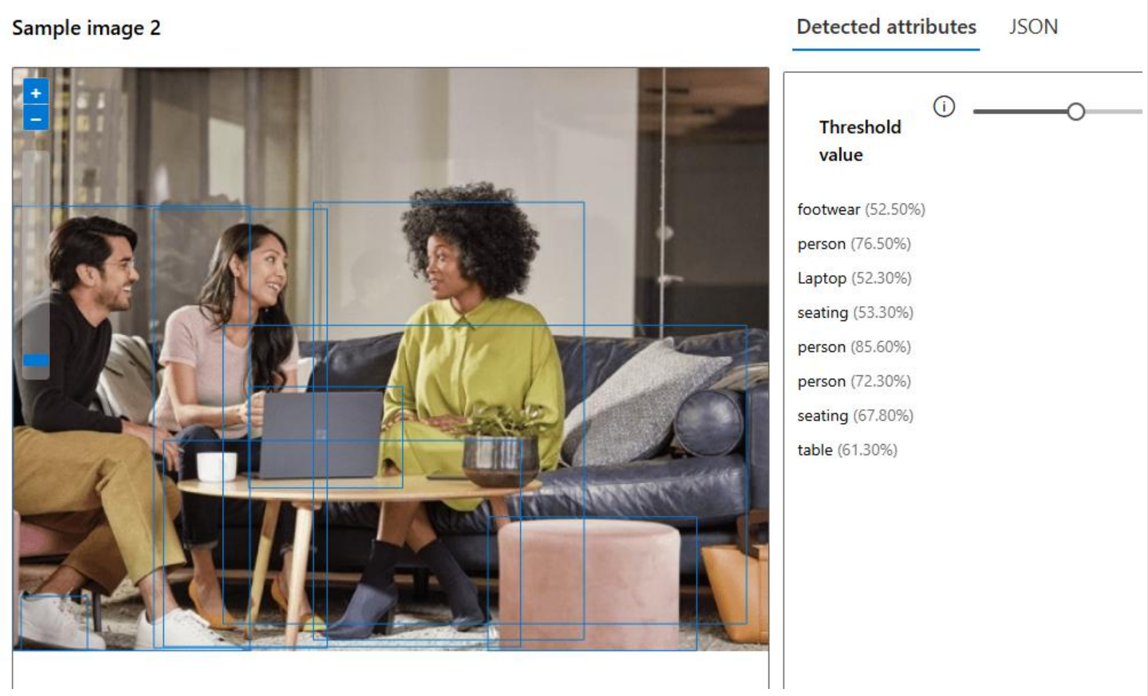
Add Captions to Images

Image captions are available through the **Caption** and **Dense Captions** features.



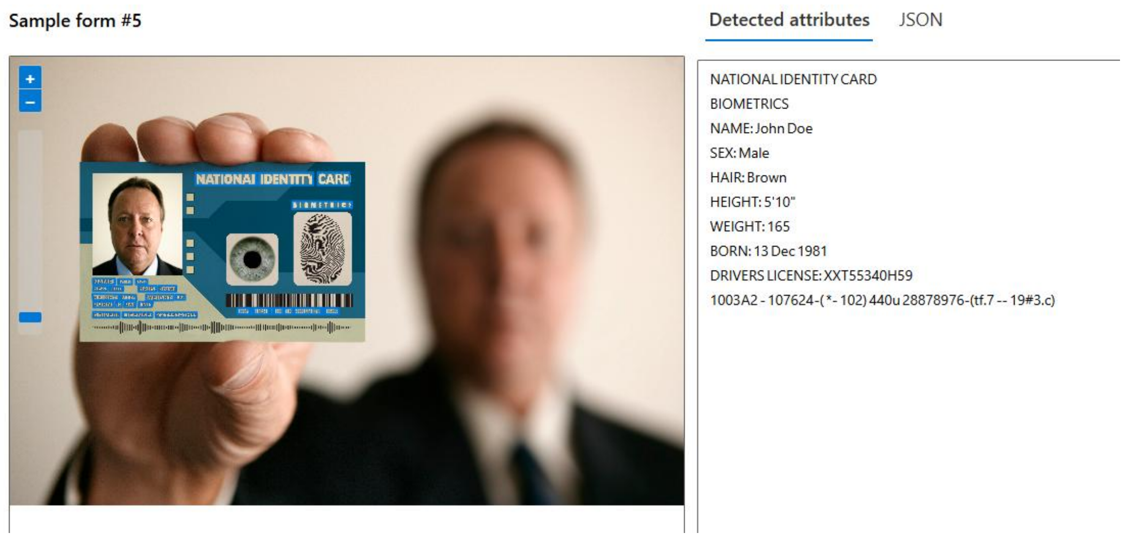
Detect Common Objects in Images

Object detection is similar to tagging, but the API returns the bounding box coordinates (in pixels) for each object found in the image



Extract Text from Images

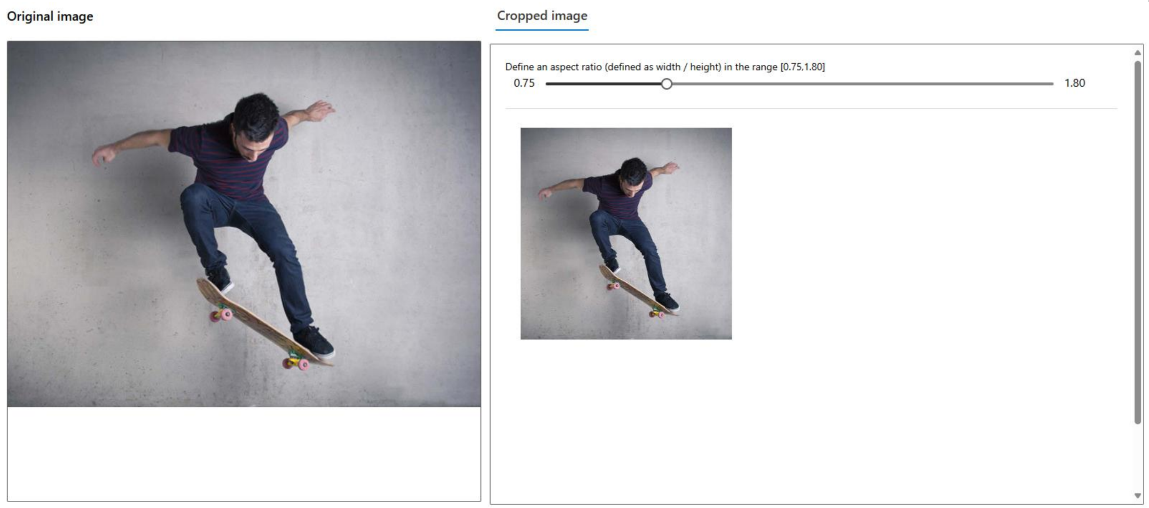
The Azure AI Vision OCR service provides a fast, synchronous API for lightweight scenarios where images aren't text-heavy.



Create Smart-cropped Images

Analyse the contents of an image to return the coordinates of the area of interest that matches a specified aspect ratio.

Computer Vision returns the bounding box coordinates of the region, so the calling application can modify the original image as desired.



People Detection

Version 4.0 of Image Analysis offers the ability to detect people appearing in images. The bounding box coordinates of each detected person are returned, along with a confidence score.

JSON


```json
{
  "modelVersion": "2024-02-01",
  "metadata": {
    "width": 300,
    "height": 231
  },
  "peopleResult": {
    "values": [
      {
        "boundingBox": {
          "x": 0,
          "y": 41,
          "w": 95,
          "h": 189
        },
        "confidence": 0.9474349617958069
      },
      {
        "boundingBox": {
          "x": 204,
          "y": 96,
          "w": 95,
          "h": 134
        },
        "confidence": 0.9470965266227722
      },
      {
        "boundingBox": {
          "x": 53,
          "y": 20,
          "w": 136,
          "h": 210
        },
        "confidence": 0.8943784832954407
      },
      {
        "boundingBox": {
          "x": 170,
          "y": 31,
          "w": 91,
          "h": 199
        },
        "confidence": 0.2713555097579956
      }
    ]
  }
}
```

Image Analysis SDK (v4.0)

- The Image Analysis SDK provides a convenient way to access the Image Analysis service using version 4.0 of the REST API.

- The SDK supports the following languages:
    - C#
    - Python
    - Java
    - JavaScript

- Please refer the following link for installation and usage instructions: Image Analysis SDK Overview - Azure AI services | Microsoft Learn

Image Analysis SDK Service Limits(v4.0)

- Image Analysis works on images that meet the following requirements:
- The image must be presented in JPEG, PNG, GIF, BMP, WEBP, ICO, TIFF, or MPO format.
- The file size of the image must be less than 20 megabytes (MB).
- The dimensions of the image must be greater than 50 x 50 pixels and less than 16,000 x 16,000 pixels.

Image Analysis API – Azure Region Availability

To use the Image Analysis APIs, you must create your Azure AI Vision resource in a supported region. The Image Analysis features are available in the following regions:

Region	Analyze Image (minus 4.0 Captions)	Analyze Image (including 4.0 Captions)	Product Recognition	Multimodal embeddings
East US	✅	✅	✅	✅
West US	✅	✅		✅
West US 2	✅		✅	✅
France Central	✅	✅		✅
North Europe	✅	✅		✅
West Europe	✅	✅		✅
Sweden Central	✅			✅
Switzerland North	✅			✅
Australia East	✅			✅
Southeast Asia	✅	✅		✅
East Asia	✅	✅		
Korea Central	✅	✅		✅
Japan East	✅			✅

Azure AI Custom Vision

Azure AI Custom Vision Overview

- Azure AI Custom Vision is an image recognition service that lets the user to build, deploy, and improve own image identifier models.

- It uses machine learning algorithms to analyse images for custom features.

- Provides two features:
  - Image classification
  - Object detection

- The service is available through
  - Custom Vision SDK/ REST API: Custom Vision SDK - Azure AI services
  - Web-based interface: https://www.customvision.ai/

Image Classification with Custom Vision

- Image Classification models can be trained in two types
  - Multilabel – Multiple tags per single image
  - Multiclass – Singe tag per image

- As a minimum it is recommend to use at least 30 images per tag in the initial image training set.

- Make sure to select the domain that matches for the image set.

- Compact domains are lightweight models that can be exported to iOS/ Android and other platforms.

Custom Vision Image Classification Domains

Domain	Purpose
Generic	Optimized for a broad range of image classification tasks. If none of the other domains are appropriate, or you're unsure of which domain to choose, select the Generic domain.
Food	Optimized for photographs of dishes as you would see them on a restaurant menu. If you want to classify photographs of individual fruits or vegetables, use the Food domain.
Landmarks	Optimized for recognizable landmarks, both natural and artificial. This domain works best when the landmark is clearly visible in the photograph. This domain works even if the landmark is slightly obstructed by people in front of it.
Retail	Optimized for images that are found in a shopping catalog or shopping website. If you want high precision classifying between dresses, pants, and shirts, use this domain.
Compact domains	Optimized for the constraints of real-time classification on mobile devices. The models generated by compact domains can be exported to run locally.

Object Detection with Custom Vision

Domain	Purpose
General	Optimized for a broad range of object detection tasks. If none of the other domains are appropriate, or if you're unsure about which domain to choose, select the General domain.
Logo	Optimized for finding brand logos in images.
Products on shelves	Optimized for detecting and classifying products on shelves.
Compact domains	Optimized for the constraints of real-time object detection on mobile devices. The models generated by compact domains can be exported to run locally.

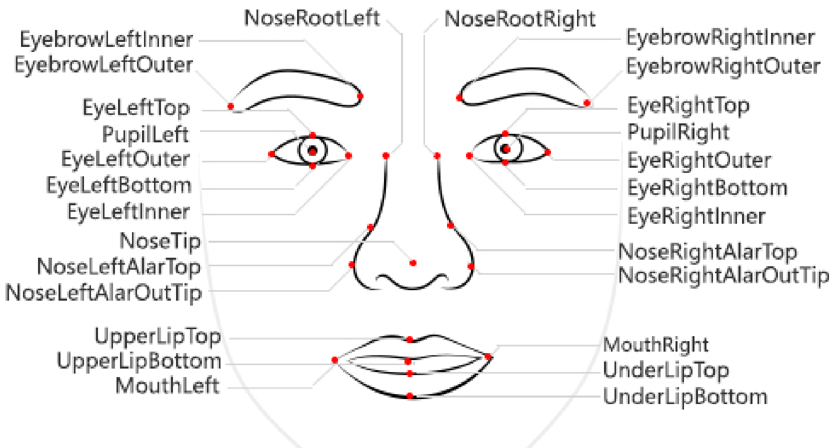
Custom Vision SDK/ Custom Vision REST API

- Azure Custom Vision SDK is available in following languages:
    - C#
    - Go
    - Java
    - JavaScript
    - Python

- Custom Vision REST API is available for training and inferencing the trained models. (Custom Vision documentation)

Azure AI Face Service

Understanding Facial Analysis



- **Face detection**
- Identifying regions of an image that contain a human face

Face analysis

facial features can be used to train ML models to return facial features such as nose, eyes, and others.

Facial recognition

Train a machine learning model to identify known individuals from their facial features.

Capabilities of Azure AI Face Service

The Azure AI Face service can return the rectangle coordinates for any human faces that are found in an image, as well as a series of related attributes:

- **Accessories**: indicates whether the given face has accessories.
- **Blur**: how blurred the face is, which can be an indication of how likely the face is to be the main focus of the image.
- **Exposure**: such as whether the image is underexposed or over exposed. This applies to the face in the image and not the overall image exposure.
- **Glasses**: whether or not the person is wearing glasses.
- **Head pose**: the face's orientation in a 3D space.
- **Mask**: indicates whether the face is wearing a mask.
- **Noise**: refers to visual noise in the image
- **Occlusion**: determines if there might be objects blocking the face in the image.
- **Quality For Recognition**: rating of high, medium, or low that reflects if the image is of sufficient quality.

Limited Access to Azure AI Face Service

Microsoft facial recognition services are Limited Access in order to help prevent the misuse of the services in accordance with Microsoft AI Principles and facial recognition principles.

- Anyone can use Face Service to:
  - Detect the **location of faces** in an image
  - Determine **if a person is wearing glasses**
  - Determine if **there's occlusion, blur, noise, or over/under exposure** for any of the faces.
  - Return the **head pose coordinates** for each face in an image.

- Require limited access policy:
  - **Face verification:** the ability to compare faces for similarity.
  - **Face identification:** the ability to identify named individuals in an image.
  - **Liveness detection:** the ability to detect and mitigate instances of recurring content and/or behaviours that indicate a violation of policies.

Limited Access Registration Process

- Customers and partners who wish to use Limited Access features of the Face API, including Face identification and Face verification, are required to register for access by submitting a registration form.
- Access to Face API is subject to Microsoft's sole discretion based on eligibility criteria and a vetting process.
- Face API is available only to customers managed by Microsoft.
- Face API is only available for certain use cases, and customers must select their desired use case in their registration.

Azure AI Vision with Responsible AI at the Core

Building AI products responsibly

Apply Responsible AI guidelines and standards throughout the software development lifecycle.

Systematically test Azure AI services for fairness and invest in diverse training data.

No images are saved to protect customer privacy.

Ensuring responsible use of AI

Limited access application for sensitive services like facial recognition and liveness validates customers, scenario, and location.

Abuse detection built into the service to help customers build safe high-quality applications.

Azure AI Video Indexer

Azure AI Video Indexer Use Cases

Deep Search

- Use the insights extracted from the video to enhance the search experience across a video library.

Content Creation

- Create trailers, highlight reels, social media content, or news clips based on the insights Azure AI Video Indexer extracts from your content.

Accessibility

- Make content available for people with disabilities or if you distribute content to different regions using different languages.

Monetization

- deliver relevant ads by using the extracted insights as additional signals to the ad server.

Content Moderation

- Use textual and visual content moderation models to keep users safe from inappropriate content and validate that the content.

Recommendations

- Video insights can be used to improve user engagement by highlighting the relevant video moments to users.

Azure AI Video Indexer

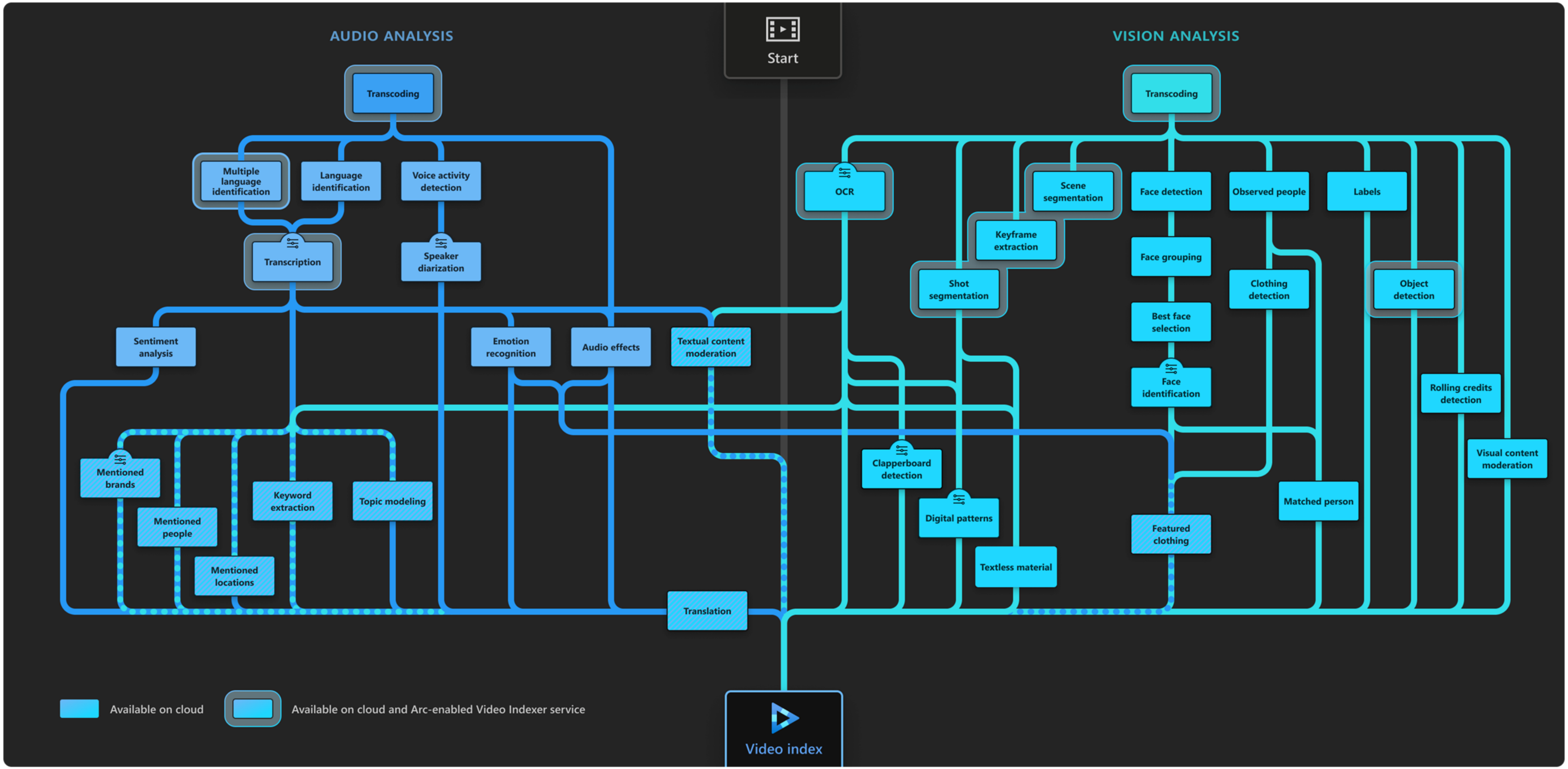
- Azure AI Video Indexer is a cloud and edge video analytics service.
- It is built on Azure AI services (such as the Face, Translator, Azure AI Vision, and Speech).
- It can extract the insights from videos using Azure AI Video Indexer video and audio models.
- No need of having machine learning expertise to retrieve actionable insights from videos.

Multichannel pipeline orchestrates visual and auditory cues and incorporates insights into a shared timeline

Run video AI indexing anywhere using Video Indexer enabled by Arc, without the need to upload your files to the cloud

Easy to evaluate and integrate, and available via web portal, web widget, and REST API

Intuitive customization to train and fine-tune selected AI models, improve content accuracy, and configure your account



Video Indexing Process

Video Models in Azure AI Video Indexer

Model	Capability
Face detection	Detects and groups faces appearing in the video.
Celebrity identification	Identifies over 1 million celebrities—like world leaders, actors, artists, athletes, researchers, business, and tech leaders across the globe.
Account-based face identification	Trains a model for a specific account. It then recognizes faces in the video based on the trained model
Thumbnail extraction for faces	Identifies the best captured face in each group of faces (based on quality, size, and frontal position) and extracts it as an image asset.
Optical character recognition (OCR)	Extracts text from images like pictures, street signs and products in media files to create insights.
Visual content moderation	Detects adult and/or racy visuals.
Labels identification	Identifies visual objects and actions displayed.
Scene segmentation	Determines when a scene changes in video based on visual cues. A scene depicts a single event and it's composed by a series of consecutive shots, which are semantically related.
Shot detection	Determines when a shot changes in video based on visual cues. A shot is a series of frames taken from the same motion-picture camera. For more information, see Scenes, shots, and keyframes.
Black frame detection	Identifies black frames presented in the video.
Keyframe extraction	Detects stable keyframes in a video.
Rolling credits	Identifies the beginning and end of the rolling credits in the end of TV shows and movies.
Editorial shot type detection	Tags shots based on their type (like wide shot, medium shot, close up, extreme close up, two shot, multiple people, outdoor and indoor, and so on). For more information, see Editorial shot type detection.

Video Models in Azure AI Video Indexer

Model	Capability
Observed people detection	Detects observed people in videos and provides information such as the location of the person in the video frame (using bounding boxes) and the exact timestamp (start, end) and confidence when a person appears.
Matched person	Matches people that were observed in the video with the corresponding faces detected. The matching between the observed people and the faces contain a confidence level.
Detected clothing	Detects the clothing types of people appearing in the video and provides information such as long or short sleeves, long or short pants and skirt or dress. The detected clothing is associated with the people wearing it and the exact timestamp (start, end) along with a confidence level for the detection are provided.
Featured clothing	Captures featured clothing images appearing in a video. You can improve your targeted ads by using the featured clothing insight. For information on how the featured clothing images are ranked and how to get the insights, see featured clothing.
Object detection	Detects unique objects that are also tracked so that if they return to the frame they are recognized.
Slate detection	Identifies the following movie post-production insights when indexing a video using the advanced indexing option: Clapperboard detection with metadata extraction, Digital patterns detection, including colour bars, Textless slate detection, including scene matching.
Textual logo detection	Matches a specific predefined text using Azure AI Video Indexer OCR. For example, if a user created a textual logo: "Microsoft", different appearances of the word Microsoft will be detected as the "Microsoft" logo.

Audio Models in Azure AI Video Indexer

Model	Capability
Audio transcription	Converts speech to text over 50 languages and allows extensions.
Automatic language detection	Identifies the dominant spoken language. For more information, see Azure AI Video Indexer language support. If the language can't be identified with confidence, Azure AI Video Indexer assumes the spoken language is English.
Multi-language speech identification and transcription	Identifies the spoken language in different segments from audio. It sends each segment of the media file to be transcribed and then combines the transcription back to one unified transcription.
Closed captioning	Creates closed captioning in three formats: VTT, TTML, SRT.
Two channel processing	Auto detects separate transcript and merges to single timeline.
Noise reduction	Clears up telephony audio or noisy recordings (based on Skype filters).
Transcript customization (CRIS)	Trains custom speech to text models to create industry-specific transcripts

Audio Models in Azure AI Video Indexer

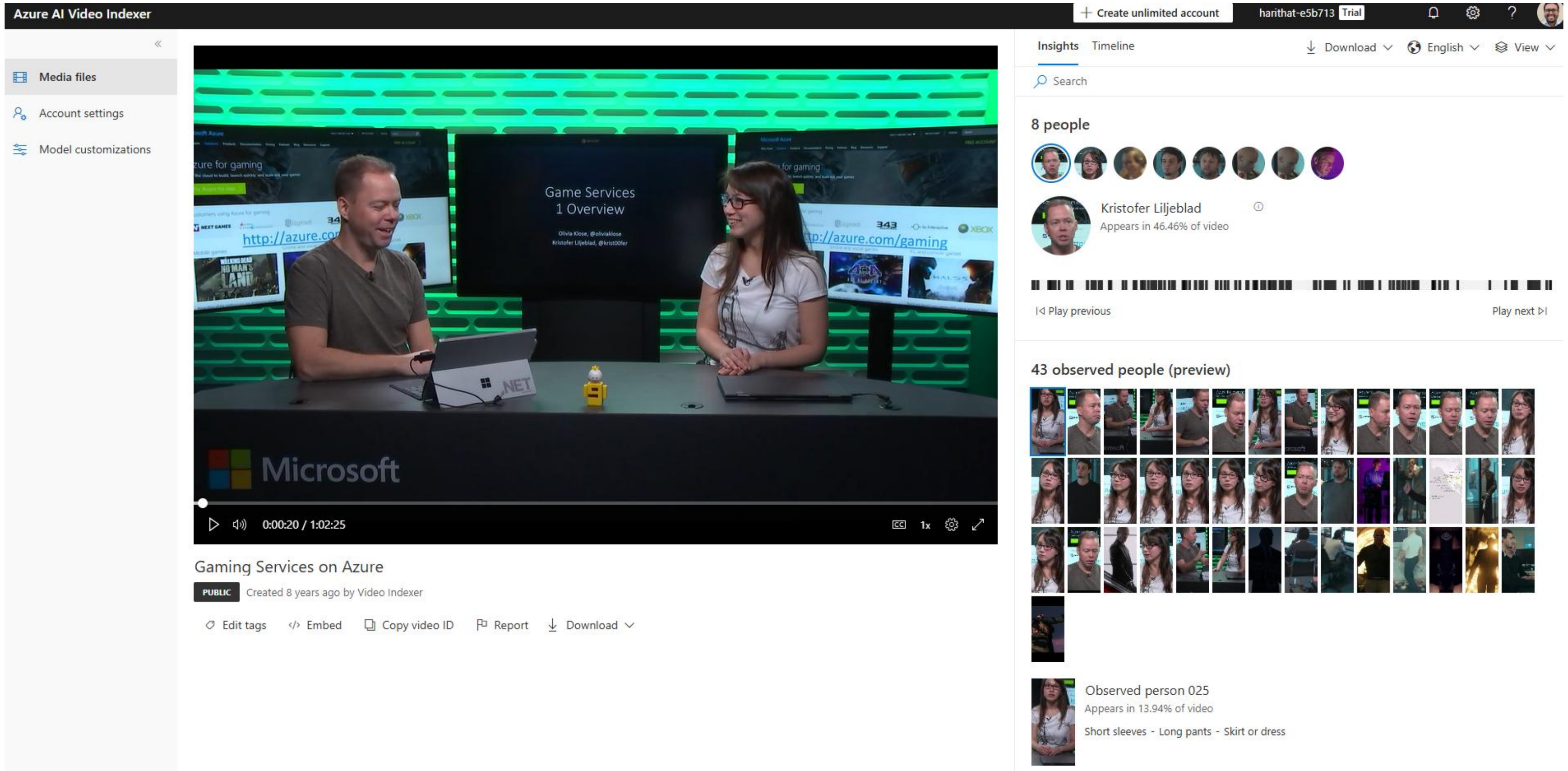
Model	Capability
Speaker enumeration	Maps and understands which speaker spoke which words and when. Sixteen speakers can be detected in a single audio-file.
Speaker statistics	Provides statistics for speakers' speech ratios.
Textual content moderation	Detects explicit text in the audio transcript.
Text-based emotion detection	Emotions such as joy, sadness, anger, and fear that were detected via transcript analysis.
Translation	Creates translations of the audio transcript to many different languages. For more information, see Azure AI Video Indexer language support.
Audio effects detection	Detects the following audio effects in the non-speech segments of the content: alarm or siren, dog barking, crowd reactions (cheering, clapping, and booing), gunshot or explosion, laughter, breaking glass, and silence.

Audio and Video Models (multi-channels)

Model	Capability
Keywords extraction	Extracts keywords from speech and visual text.
Named entities extraction	Extracts brands, locations, and people from speech and visual text via natural language processing (NLP).
Topic inference	Extracts topics based on various keywords (e.g., keywords 'Stock Exchange', 'Wall Street' produce the topic 'Economics'). The model uses three different ontologies (IPTC, Wikipedia, and the Video Indexer hierarchical topic ontology). The model uses transcription (spoken words), OCR content (visual text), and celebrities recognized in the video using the Video Indexer facial recognition model.
Artifacts	Extracts a rich set of "next level of details" artifacts for each of the models.
Sentiment analysis	Identifies positive, negative, and neutral sentiments from speech and visual text.

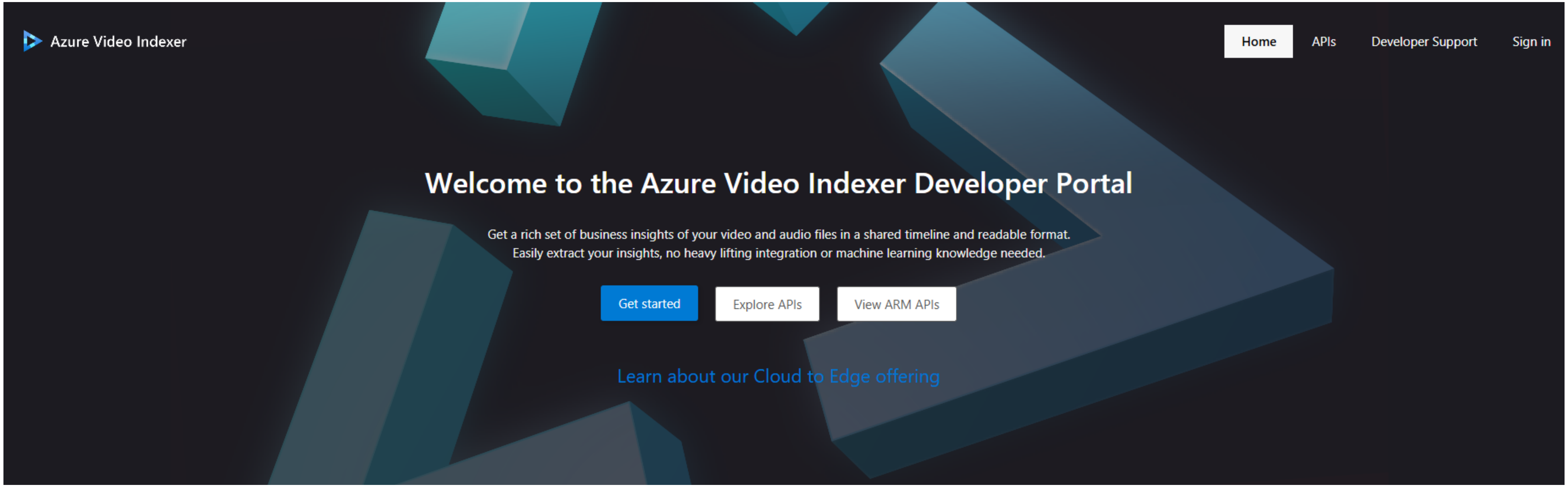
Azure AI Video Indexer Web Portal

http://www.videoindexer.ai



Azure AI Video Indexer Developer Portal

http://api-portal.videoindexer.ai



Azure AI Video Indexer Resources

Video Indexer API	https://api-portal.videoindexer.ai/
Docs	https://learn.microsoft.com/en-us/azure/azure-video-indexer/
Video Indexer Web Portal	www.videoindexer.ai
Samples Repositories	https://github.com/Azure-Samples/azure-video-indexer-samples

Azure AI Video Indexer enabled by Arc (Preview)

- Azure AI Video Indexer enabled by Arc is an Azure Arc extension enabled service that runs video and audio analysis, and generative AI on edge devices.
- The solution is designed to run on Azure Arc enabled Kubernetes
- The Azure AI Video Indexer extension installs and deploys Azure AI Video indexer to the Kubernetes cluster.
- It supports many video formats, including MP4 and other common formats.
- View the updated documentation on: What is Azure AI Video Indexer enabled by Arc?

Demonstration

- Optical Character Recognition (OCR)
- Image Analysis
- Face
- Azure AI Video Indexer

Lab – Azure AI Vision

Lab Description

Azure AI Vision is a cloud-based service from Microsoft that uses advanced algorithms to analyze images and extract valuable information. It includes capabilities like Optical Character Recognition (OCR), face detection, image analysis, and video indexing. This lab guides you through low-code exercises. You also have the opportunity to work on pro-code lab

Questions?

Thank you